Unit 3, Lesson 2: Writing Large Numbers Using Scientific Notation

Benchmark

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Learning Targets

- ☐ **I can express large numbers using scientific notation.**
- ☐ **I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.**

3.2.1 Warm-Up: Googol, not Google

A "googol" is a name for a really big number: a 1 followed by 100 zeros.

1. If you square a googol, how many zeros will the answer have? Show your reasoning.

3.2.2 Exploration: Patterns in Scientific Notation

Very large numbers can be written using powers of 10 in a shorthand method called scientific notation.

1. Work with your partner to complete the table.

Power of 10	Value	Scientific Notation	Word Form
10^2	100	1.0×10^2	one hundred
10^3		1.0×10^3	
		$1.0 \times 10^{\boxed{}}$	ten thousand
	100,000	$\boxed{} \times 10^5$	
10^6		$\boxed{} \times 10^{\boxed{}}$	
	1,000,000,000		
		1.0×10^{12}	one trillion

2. Select two of the numbers. Work with your partner to name something in the world that is described by those numbers.

Number	What can be described by this number in the real world?

3. Work with your partner to describe at least two patterns in the relationships among the power of 10, the numerical value, the scientific notation, and/or the word form.

a.

b.

3.2.3 Guided Instruction: Writing Large Numbers in Scientific Notation

1. **Consider the number 2,000,000.**

a. **What power of ten represents 1 million?** $10^{\square}$

b. **Complete the multiplication sentence to express 2,000,000 using this power of 10.**

$\square \times 10^{\square} = 2{,}000{,}000$

Scientific notation is a method of writing very large or very small numbers using exponents in which a number is expressed as the product of a power of 10 and a number that is at least 1 and less than 10.

2. **Complete the statement.**

When rewriting 2,000,000 in scientific notation, the factor 2 is

- o less than
- o greater than or equal to

1 and

- o less than
- o greater than or equal to

10. The digit 2 is in the millions place. A million expressed as a power of 10 is

- o 10^5.
- o 10^6.

Therefore, 2,000,000 is written as

- o 2×10^5
- o 2×10^6

in scientific notation.

3. **The expression 20×10^5 is also equivalent to 2,000,000. Use the definition of scientific notation to explain why 20×10^5 is not the correct expression of 2,000,000 in scientific notation.**

4. Consider the standard form and scientific notation of the numbers shown in the table.

Standard Form	Scientific Notation
2,300,000	2.3×10^6
2,030,000	2.03×10^6

Describe the similarities and differences between the values expressed in scientific notation.

Similarities	Differences

As explored in the previous lesson, the powers of 10 relate directly to place values.

5. The planet Venus is about 26,000,000 miles from earth.

a. What value between 1 and 10 could be multiplied by a power of 10 to result in 26,000,000?

b. Which digit is in the ones place of your answer to Part A?

c. In which place value is that digit in 26,000,000?

d. Which power of 10 relates to this place value?

e. Express 26,000,000 in scientific notation.

6. In 2018, funding for education in Florida totaled $30,100,000,000. Express this value in scientific notation.

3.2.4 Guided Practice: Writing Large Numbers in Scientific Notation

1. The table shows the number of visitors to several amusement parks in Florida in 2019. Express each value in scientific notation.

Florida Theme Park	Approximate Number of Visitors in 2019	Equivalent Value in Scientific Notation
Magic Kingdom	20,960,000	
Animal Kingdom	13,900,000	
Busch Gardens	4,200,000	
Sea World	4,640,000	
Adventure Island	656,000	

3.2.5 Collaboration: Comparing Large Numbers Written in Scientific Notation

1. Work with your partner to complete the following.

a. Complete the table.

Scientific Notation	Standard Form
	200
2×10^3	
2×10^4	
	200,000

b. Complete each statement.

2×10^3 is
- o 10
- o 100
- o 1000

times
- o smaller
- o larger

than 2×10^2.

2×10^2 is
- o 10
- o 100
- o 1000

times
- o smaller
- o larger

than 2×10^4.

2×10^5 is
- o 10
- o 100
- o 1000

times
- o smaller
- o larger

than 2×10^2.

c. Describe how to compare numbers written in scientific notation if the factors being multiplied by a power of 10 are the same.

2. Complete the table.

Scientific Notation	Standard Form
	32,000,000
6.4×10^7	
	96,000,000

a. How many times larger is 6.4×10^7 than 3.2×10^7?

b. How many times smaller is 3.2×10^7 than 9.6×10^7?

c. Describe how to compare numbers written in scientific notation if the powers of 10 are the same.

3. Complete the table.

Scientific Notation	Standard Form
2.1×10^4	
	420,000
8.4×10^6	

a. How many times larger is 4.2 than 2.1? $\boxed{}$

b. How many times larger is 10^5 than 10^4? $\boxed{}$

c. How many times larger is 4.2×10^5 than 2.1×10^4?

$\boxed{} \times \boxed{} =$ ________ times larger

d. Complete the statement.

The value of 8.4×10^6 is

- o 4
- o 40
- o 400

times larger than the value of 2.1×10^4 because 8.4 is

- o 4
- o 40
- o 400

times larger than 2.1, 10^6 is

- o 10
- o 10^2
- o 10^3

times larger than 10^4, and [ ] x [ ] =

- o 4.
- o 40.
- o 400.

If the factors being multiplied by the powers of 10 are different, but one is not a multiple of the other, the comparison of the two values can be *approximated*.

4. The average distance from Earth to the moon is 2.38855×10^5 miles. The average distance from Earth to the sun is about 9.3×10^7.

   Complete the statements to approximate how many times farther the sun is from Earth than the moon.

   9.3 is about

   - o 2
   - o 4
   - o 6

   times

   - o smaller
   - o larger

   than 2.38855 and 10^7 is

   - o 10
   - o 10^2
   - o 10^3

   times

   - o smaller
   - o larger

   than 10^5, so the value of 9.3×10^7 is about ___________ times larger than 2.38855×10^5.

3.2.6 Your Turn!

1. Each statement contains a quantity. Rewrite each quantity in scientific notation.

   a. There are about 37,000,000,000 cells in the average human body.

   b. Light travels through space at a rate of 300,000,000 meters per second.

2. The population of Florida was about 2.22×10^7 in 2021. The population of the world was about 7.87×10^9 in the same year.

   Complete the statement.

   In 2021, the population of the world was about ___________ times larger than the population of Florida.

3.2.7 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate where your current learning is related to each target.

   a. I can express large numbers using scientific notation.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

   b. I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

